Ana Trišović

Computer Science & Artificial Intelligence Laboratory (CSAIL)
The Institute for Quantitative Social Science (IQSS) at Harvard
anatrisovic.com | ana_tris@mit.edu

Research Interests

I measure how AI is changing science: how fast it diffuses across disciplines, what drives its uptake, and how scientists actually use models, including how quickly dependence on specific models forms and decays. This builds on over a decade of work on reproducibility, from data provenance at CERN to large-scale studies of research code.

Professional Experience

Sep 2023 – Present
Massachusetts Institute of Technology
- *Research Scientist* at FutureTech - Lead research on the societal and scientific impact of AI across disciplines, industries, and geopolitical contexts. PI of the first Sloan Foundation grant at CSAIL.

Sep 2019 – Present
Harvard University
- *Research Associate* (Feb 2022 – Present) at The Institute for Quantitative Social Science
  Advance data infrastructure and user research to optimize the Dataverse platform's capabilities.
- *Research Associate* (Feb 2022 – Sep 2023) at Harvard T.H. Chan School of Public Health
  Built climate–health data pipelines. Co-authored NIH grants securing over $6M.
- *Postdoctoral Fellow* (Sep 2019 – Feb 2022) at The Institute for Quantitative Social Science
  Advanced Dataverse repository, built partnerships, and wrote a funded Sloan Foundation proposal.

Sep 2018 – Sep 2019
University of Chicago
- *CLIR Postdoctoral Fellow* at The Energy Policy Institute - Supported modeling of climate policy impacts, including estimating the social cost of carbon using large-scale environmental data.

Jul 2013 – Aug 2018
CERN
- *Project Associate* (Sep 2017 – Aug 2018) – Contributed to CERN Open Data and Analysis Preservation platforms, advancing infrastructure for reproducible and accessible research.
- *Doctoral Student* (Aug 2014 – Sep 2017) – Designed systems to capture dependencies between software, detectors, and datasets for LHCb, supporting seamless data analysis across upgrades.
- *Technical Student* (Jul 2013 – Jul 2014) – Developed a particle collision display tool for LHCb, now used by students worldwide through the International Masterclass program.

Mar 2013 – Jul 2013
Microsoft
- *Data Science Associate* - Analyzed Azure cloud metrics to assess load balancing and blacklisting systems, proposing optimizations for performance and efficiency.

Education

2014 - 2018 **University of Cambridge, Newnham College**
Ph.D. in Computer Science
Thesis title: Data preservation and reproducibility at the LHCb experiment at CERN
Advisers: Christopher Jones, Benjamin Couturier
Committee: Richard Batley, Andrew McNab

2010 - 2014 **Union University, School of Computing** in Serbia
B.Sc. in Computer Science

2010 - 2013 **University of Belgrade** in Serbia
B.Sc. in Mechanical Engineering

Selected Awards

2025	Excelencia Severo Ochoa Mobility Fellow
2025	NSF SeekCommons Fellow
2025	Breakthrough Prize as part of LHCb Collaboration
2019	Research Data Alliance (RDA) Early-Career Fellowship
2018	CLIR Postdoctoral Fellowship, Council on Library and Information Resources
2016	Computing in High Energy Physics (CHEP) Student Scholarships
2015–2017	Muir Wood Scholarship, Newnham College, University of Cambridge
2014	Google Anita Borg Memorial Scholarship (Women Techmakers)
2013–2015	Dositeja Award - for extraordinary students in Serbia.
2012	Deutscher Akademischer Austauschdienst (DAAD) Scholarship

Publications

†: equal contributions. ‡: corresponding / senior authorship. underline: student / mentee

1. **Ana Trišović**. "The Shrinking Lifespan of LLMs in Science". *Conference on Language Modeling (COLM)* (2026)
2. **Ana Trišović** and Janakan Sivaloganathan. "Open Science, Closed Models: How Funding Shapes AI in Science". *AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES)* (2026)
3. Hans Gundlach, Alex Fogelson, Jayson Lynch, **Ana Trišović**, Jonathan S. Rosenfeld, Anmol Sandhu, and Neil Thompson. "On the Origin of Algorithmic Progress in AI". *Conference on Neural Information Processing Systems (NeurIPS)* (2026)
4. **Ana Trišović**, Jan Range, Philip Durbin, Katherine Mika, Amber Leahey, and Wei Li. "Advancing Geospatial Data Infrastructure in Dataverse via Metadata Automation, Interactive Tools and LLM Case Study". *Environmental Modelling & Software* (2025)
5. Jie Kate Hu†, **Ana Trišović**†, Ankita Bakshi, Danielle Braun, Francesca Dominici, and Joan Casey. "Co-exposure to Extreme Heat, Wildfire Burn Zones, and Wildfire Smoke in the Western US from 2006 to 2020". *Science Advances* (2025)
6. James A. Rising, Azhar Hussain, Kevin Schwarzwald, and **Ana Trišović**. "A Practical Guide to Climate Econometrics: Navigating Key Decision Point in Weather and Climate Data Analysis". *Journal of Open Source Education (JOSE)* (2025)
7. Janakan Sivaloganathan, **Ana Trišović**, and Neil Thompson. "Mapping the Impact of Foundation Models on the UN Sustainable Development Goals". *21st IEEE International Conference on e-Science* (2025)
8. Alex Fogelson, **Ana Trišović**, and Neil Thompson. "LLMs in Citation Intent Classification: Progress, Precision, and Reproducibility Challenges". *ACM Conference on Reproducibility and Replicability (ACM REP)* (2025)
9. **Ana Trišović**, Jan Range, Philip Durbin, Amber Leahey, Wei Li, and Danielle Braun. "Improving FAIR Compliance for High-Dimensional Data via Automated Metadata Extraction". *2025 IEEE International Conference on eScience (eScience)* (2025)
10. Mauricio Tec, **Ana Trišović**, Michelle Audirac, Sophie Woodward, Jie Kate Hu, Naeem Khoshnevis, and Francesca Dominici. "`SpaCE`: The Spatial Confounding Environment". *12th International Conference on Learning Representations (ICLR)* (2024). Earlier version at CLeaR (Causal Learning and Reasoning) 2023

11. Whanhee Lee, Xiao Wu, Seulkee Heo, Joyce Mary Kim, Kelvin C Fong, Ji-Young Son, Matthew Benjamin Sabath, **Ana Trišović**, Danielle Braun, Jae Yoon Park, Yong Chul Kim, Jung Pyo Lee, Joel Schwartz, Ho Kim, Francesca Dominici, Ziyad Al-Aly, and Michelle L. Bell. "Air Pollution and Acute Kidney Injury in the US Medicare Population: A Longitudinal Cohort Study". *Environmental Health Perspectives* (2023)

12. **Ana Trišović**, Matthew K. Lau, Thomas Pasquier, and Mercè Crosas. "A Large-scale Study on Research Code Quality and Execution". *Nature Scientific Data* (2022). Covered in: The Scientist, ReproducibiliTea Podcast

13. Daniel Garijo, Hervé Ménager, Lorraine Hwang, **Ana Trišović**, Michael Hucka, Thomas Morrell, Alice Allen, Task Force on Best Practices for Software Registries, and SciCodes Consortium. "Nine Best Practices for Research Software Registries and Repositories". *PeerJ Computer Science* (2022)

14. Layan Bahaidarah, Ethan Hung, Andreas Francisco De Melo Oliveira, Jyotsna Penumaka, Lukas Rosario, and **Ana Trišović**[‡]. "Toward reusable science with readable code and reproducibility". *18th International Conference on e-Science (e-Science)* (2022)

15. Stian Soiland-Reyes, Peter Sefton, Mercè Crosas, Leyla Jael Castro, Frederik Coppens, José M Fernández, Daniel Garijo, Björn Grüning, Marco La Rosa, Simone Leo, Eoghan Carragáin, Marc Portier, **Ana Trišović**, RO-Crate Community, Paul Groth, and Carole Goble. "Packaging research artefacts with RO-Crate". *Data Science* (2021)

16. **Ana Trišović**, Katherine Mika, Ceilyn Boyd, Sebastian Feger, and Mercè Crosas. "Repository Approaches to Improving the Quality of Shared Data and Code". *Data* (2021)

17. Aleksandrina Goeva, Sara Stoudt, and **Ana Trišović**. "Toward Reproducible and Extensible Research: from Values to Action". *Harvard Data Science Review* (2020)

18. **Ana Trišović**, Chris R. Jones, Ben Couturier, and Marco Clemencic. "Provenance Tracking in the LHCb Software". *Computing in Science & Engineering (CISE)* (2020)

19. Anna E. Woodard, **Ana Trišović**, Zhuozhao Li, Yadu Babuji, Ryan Chard, Tyler Skluzacek, Ben Blaiszik, Daniel S. Katz, Ian Foster, and Kyle Chard. "Real-Time HEP Analysis With FuncX – a High-Performance Platform for Function as a Service". *24th International Conference on Computing in High Energy & Nuclear Physics (CHEP)* (2020)

20. Xiaoli Chen, Sünje Dallmeier-Tiessen, Robin Dasler, Sebastian Feger, Pamfilos Fokianos, Jose Benito Gonzalez, Harri Hirvonsalo, Dinos Kousidis, Artemis Lavasa, Salvatore Mele, Diego R. Rodriguez, Tibor Simko, Tim Smith, **Ana Trišović**[‡], Anna Trzcinska, Ioannis Tsanaktsidis, Markus Zimmermann, Kyle Cranmer, Lukas Heinrich, Gordon Watts, Michael Hildreth, Lara Lloret Iglesias, Kati Lassila-Perini, and Sebastian Neubert. "Open Is Not Enough". *Nature Physics* (2019)

21. Thomas Pasquier, Matthew K. Lau, **Ana Trišović**, Emery R. Boose, Ben Couturier, Mercè Crosas, Aaron M Ellison, Valerie Gibson, Chris R. Jones, and Margo Seltzer. "If These Data Could Talk". *Nature Scientific Data* (2017)

22. **Ana Trišović**, Ben Couturier, Val Gibson, and Chris R. Jones. "Recording the LHCb Data and Software Dependencies". *22nd International Conference on Computing in High Energy & Nuclear Physics (CHEP)* (2017)

Peer-reviewed Workshop Papers

23. **Ana Trišović**[†], Alex Fogelson[†], Janakan Sivaloganathan, and Neil Thompson. "Science Is Falling Behind the Frontier: Foundation Model Adoption Across Half a Million Papers". *AI4MetaScience: AI for Meta-Science Workshop (NeurIPS 2026)* (2026)

24. Kristina Sakayeva and **Ana Trišović**‡. "Reproducing ELEPHANT: Judge-Dependence and Model Deprecation in an LLM Sycophancy Benchmark". *TAE (Trust-AI-Eval): Can We Trust AI Evaluation? Workshop and AI4MetaScience: AI for Meta-Science Workshop (NeurIPS 2026)* (2026). Under review at Transactions on Machine Learning Research (TMLR)

25. F. Martino O'Rourke, **Ana Trišović**, and Dimitris Bertsimas. "A Causal DAG Prior for Synthetic Time-Series Classification Datasets". *Workshop on Foundation Models for Structured Data at ICML* (2026)

26. **Ana Trišović**, Gabe Miller, Dimitris Bertsimas, and Jie K. Hu. "Predicting Concurrence of Heatwaves, Droughts, and Wildfires with Spatiotemporal Deep Learning". *Tackling Climate Change with Machine Learning at ICLR* (2025)

27. **Ana Trišović**, Philip Durbin, Tania Schlatter, Gustavo Durand, Sonia Barbosa, Danny Brooke, and Mercè Crosas. "Advancing Computational Reproducibility in the Dataverse Data Repository Platform". *3rd International Workshop on Practical Reproducible Evaluation of Computer Systems (P-RECS)* (2020)

Preprints and Work in Progress

28. M. Codreanu, A. Imas, J. Mateos-Garcia, J. Emmens, E. Muiruri, A. Turrell, J. Jacobs, A. Kasirzadeh, **Ana Trišović**, Y. Chen, T. Rodchenko, C. Pollard, S. Strand, D. Rock, Z. Iscenko, F. Curto Millet, N. Thompson, and J. Manyika. "AI in Science: Early Insights" (2026)

29. **Ana Trišović**, Alex Fogelson, and Neil Thompson. "Four Facts About US-China AI Competition in Science". *Available at SSRN 6028854* (2025). Covered in: MIT Science Policy Review

30. Romain Puech, Joseph Ye, Danique de Moor, **Ana Trišović**, and Dimitris Bertsimas. "Rapid drone-based wildfire detection at a fraction of current prevention spending". *Submitted to Nature Communications*

Research Grants

2025–2027 **Alfred P. Sloan Foundation** ($300,000)
Principal Investigator (co-developed and conceptualized the proposal) with Dr. Neil Thompson as Co-PI, for quantifying the use of AI in science and exploring automation in science.

2025–2026 **MIT Generative AI Impact Consortium** ($150,000)
Key Personnel (co-conceived and co-wrote the funded proposal) with Prof. Dimitris Bertsimas as PI, "Personalized Learning with GenAI for MIT Open Learning Courseware".

2023–2025 **Microsoft - Innovation + Society** ($300,000)
Key Personnel (co-conceived and co-wrote the funded proposal) with Dr. Neil Thompson as PI, "Measuring the accessibility and democratization of the AI ecosystem in science".

2024–2026 **OpenAI Credits for Research** ($11,000 total)
Project Lead (proposal concept & development) for large-scale bibliometric analysis of AI in science.

2023–2026 **National Institute of Environmental Health Sciences** ($6.7 million)
Key Personnel (co-conceived and co-wrote the funded proposal) with Prof. Francesca Dominici as PI and others for creating a Research Coordinating Center (RCC) on climate change and health

2022-2024 **Institute of Museum and Library Services** ($99,372)
PI (assumed the PI role after Dr. Daina Bouquin's departure), for Software Citation Implementation: Action Plan Development.

2022–2023 **National Institutes of Health** ($238,197)
Key Personnel (conceived and wrote the funded proposal), for supporting dissemination of geospatial data in the Dataverse repository.

2022–2023 **National Institutes of Health** ($336,874)

Key Personnel (co-conceived and co-wrote the funded proposal), for developing approaches for AI/ML readiness for wildfire exposures.

2020–2023 **Alfred P. Sloan Foundation** ($196,990)

Key Personnel; co-conceived and co-wrote the funded proposal. PIs Prof. Gary King and Dr. Mercè Crosas (Prof. King sole PI after Dr. Crosas's departure), "Preservation of advanced computing artifacts and automatic code revision for research reproducibility and reuse".

2019–2021 **CLIR Micro-Grant for Research** ($3,000)

Key Personnel (conceived and wrote the funded proposal), for exploring gamification in research dissemination.

2019–2020 **Amazon AWS Cloud Credits for Research** ($10,000)

Project Lead (conceived and wrote the funded proposal), for large-scale research software analysis in Dataverse.

Teaching Experience

2025 **Universal AI (UAI) at MIT Open Learning**
Course creator and instructor in Data Analytics and Machine Learning.

2024 **NASA's Transform to Open Science Summer School, UIUC**
Lecturer on Open Science and AI methods.

2023 **Clinical Research Education in Genome Science (CREiGS), School of Medicine at Mount Sinai**
Lecturer on Reproducibility in Scientific Computing.

2023 **International Summer School on Deep Learning & Big Data (IRDTA)**
Lecturer on Principles of Computational Research.

2021 **Summer School at the Xidian University in China**
Lecturer in Introduction to Machine Learning.

2019 **Energy Policy Institute at UChicago**
Lecturer in Introduction to Geospatial Analysis.

2017 **LHCb Experiment at CERN, Switzerland**
Instructor in Fundamentals of Data Analysis in Particle Physics.

2015 **Newnham College, University of Cambridge**
TA in Algorithms (Undergraduate).

2015 **Cavendish Laboratory, University of Cambridge**
TA in Physics Practicals (Undergraduate) and Computational Physics (Graduate).

Research Advising & Mentoring

2026 Franco Martino O'Rourke MBA '26. MIT.
2026 Romain Puech PhD '27. MIT.
2025 Caio de Prospero Iglesias MBA '25. MIT.
2025 Gabe Miller MBA '25. MIT.
2024 Anna Tsvetkov PhD '25. Brown University.
2022 Lauren Flynn M.Sc. '23. Harvard T.H. Chan School of Public Health.
2022 Ivan Gu M.Sc. '22. Harvard Medical School.
2022 Nathaniel D. Fairbank Intern.
2020 - 2021 Layan Bahaidarah B.Sc. '20. Boston University.
2020 - 2021 Ethan Hung B.Sc. '20. Boston University.

2020 - 2021 Andreas F. De Melo Oliveira B.Sc. '20. Boston University.
2020 - 2021 Jyotsna Penumaka B.Sc. '20. Boston University.
2020 - 2021 Lukas Rosario B.Sc. '20. Boston University.

Research Assistants

2024 - 2026 Alex Fogelson
2023 - 2026 Janakan Sivaloganathan

Undergraduate Research Assistants at MIT

2025 - PRESENT Kristina Sakayeva
2025 - PRESENT Maeve Zimmer
2026 Adem Bizid
2025 Ingrid Tomovski
2024 - 2025 Selinna Lin
2024 - 2025 Evan Zhang
2024 - 2025 Grace Yuan
2024 - 2025 Sri Saraf
2024 - 2025 Denis-Bogdan Siminiuc
2025 Joseph Ye
2025 Emma Li
2025 Yibo Cheng
2025 Alvin Banh
2024 Elizabeth Yan
2024 Dora M. Zhou
2024 Raul D. Campos

Invited talks, Keynotes and Panels

2026 **Invited talk**, Data Science and AI for Public Good Webinar, Swiss Federal Statistical Office
2026 **Keynote**, StartSe AI Festival, São Paulo, Brazil
2025 **Invited talk**, 650th EUROMECH Colloquium, Belgrade, Serbia
2025 **Invited talk**, 21st IEEE International Conference on eScience, Chicago, IL
2025 **Invited talk**, Barcelona Supercomputing Center (BSC), Barcelona, Spain
2025 **Panel moderator**, Special Closed Session: "Defining the Parameters of Success for a National Next-Gen AI Initiative," Government-University-Industry-Philanthropy Research (GUIPRR), National Academies of Sciences, Engineering, and Medicine (NASEM), Washington, DC
2025 **Invited talk**, IAP Seminar, MIT Operations Research Center (ORC), Cambridge, MA
2024 **Invited talk**, OpenForum Academy Symposium, Cambridge, MA
2024 **Invited talk**, Annual Conference of the MIT Initiative on the Digital Economy (IDE), Cambridge, MA
2023 **Invited talk**, Cultivarium, Watertown, MA
2022 **Panelist**, Birds of a Feather (BoF) Session, Practice and Experience in Advanced Research Computing (PEARC22), Boston, MA
2022 **Panelist**, International Conference on Supercomputing (SC22), Dallas, TX
2022 **Invited talk**, Statistical Software Conference, University of Toronto, Toronto, Canada

2022 **Invited talk**, Journal Editors Discussion Interface (JEDI) Workshop, Online
2022 **Invited talk**, Mathematical Institute of the Serbian Academy of Sciences and Arts, Belgrade, Serbia
2022 **Invited talk**, Pew Research Center, Washington, DC
2021 **Invited talk**, Gathering for Open-Source Hardware (GOSH) Meeting, Online
2021 **Invited talk**, Helmholtz Rich Metadata Software Publication (HERMES) Workshop, Online
2021 **Invited talk**, Metascience Conference, Online
2021 **Invited talk**, Expert Meeting on Data Sharing Guidance for NIA Longitudinal Studies, National Academies (CNSTAT), Online
2019 **Keynote**, Building Reproducible Workflows for Earth Sciences Workshop, Reading, UK
2016 **Invited talk**, Multidisciplinary Perspectives on Data Preservation and Access Workshop, Cambridge, UK

Academic Service

2024 – Present **Member**, U.S. National Committee for CODATA, National Academies (NASEM)
2021 – 2023 **Committee Member**, Consortium of Scientific Software Registries and Repositories (SciCodes)
2021 – 2023 **Chair**, GDCC Software and Container Metadata Working Group

Conference Service

2026 **Area Chair**, AI-Native Academia @ NeurIPS 2026
2025 **Organization Committee**, ACM Conference on Reproducibility and Replicability
2025 **Workshop Organization Committee**, Academy of Management (AOM)
2024 **Reproducibility Initiative Committee Member**, Supercomputing Conference
2023 **Organization Committee**, IEEE International Conference on eScience
2021 **Organizer**, CLIR Webinar: Toward Open, Reproducible and Reusable Research
2020 – 2021 **Planning Committee and Chair**, Annual Dataverse Community Meeting
2020 **Program Committee**, SciPy Scientific Computing Conference
2020 **Program Committee**, The Theory and Practice of Provenance (TaPP)

Editorial Service

2023 – Present **Topic Editor**, Journal of Open-Source Software (JOSS)
2021 – Present **Review Board Member**, Journal of Systems Research (JSys)

Grant Reviewer

National Science Foundation
The Sloan Foundation